

IQUIQUE, Chile

WMO: 854180

Lat: 20.5397S

Lon: 70.1786W

Elev: 48

StdP: 100.75

Time Zone: -4.00 (CHL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.1	12.9	6.9	6.2	13.8	7.8	6.6	14.4	10.0	16.8	9.1	16.7	2.5	160	0.413

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	5.8	26.8	20.1	25.9	19.6	25.1	19.2	21.0	25.2	20.4	24.6	19.8	24.1	6.8	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.2	14.0	23.9	18.6	13.5	23.4	18.0	13.0	23.0	61.1	25.0	58.5	24.8	56.7	24.2	27.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.9	9.0	8.2	10.5	28.4	1.0	1.0	9.8	29.2	9.2	29.8	8.6	30.4	7.9	31.2		
			DB	8.1	21.8	1.3	1.7	7.2	23.0	6.5	24.0	5.7	24.9	4.8	26.2	
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.5	21.9	22.3	21.2	19.3	17.7	16.4	15.6	15.6	16.2	17.2	18.6	20.3
	DBStd	2.67	1.39	1.36	1.38	1.43	1.55	1.28	1.25	1.19	1.20	1.19	1.06	1.23
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	380	0	0	1	6	32	60	84	84	64	38	9	1
	CDD10.0	3106	370	344	348	278	238	191	175	175	187	223	257	320
	CDD18.3	443	112	111	90	34	11	1	1	1	1	3	17	62
	CDH23.3	822	267	325	166	25	3	0	0	2	0	0	0	34
	CDH26.7	24	7	13	4	0	0	0	0	1	0	0	0	0
Wind	WSAvg	3.8	4.4	4.6	4.3	3.8	3.3	3.1	3.2	3.3	3.6	3.8	3.9	4.2
Precipitation	PrecAvg	6	0	0	0	0	0	0	1	1	4	0	0	0
	PrecMax	87	0	0	0	0	0	0	13	9	78	0	0	1
	PrecMin	0	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	19	0	0	0	0	0	0	3	2	17	0	0	0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.4	28.1	27.1	25.2	23.2	21.0	20.1	20.2	20.7	21.2	22.8	25.2
		MCWB	20.8	20.2	20.1	19.0	17.8	16.2	15.5	16.5	16.5	15.8	16.9	19.2
	2%	DB	26.2	26.9	26.0	24.0	22.1	20.0	19.0	19.0	19.2	20.2	21.9	24.2
		MCWB	20.2	19.9	19.6	18.4	17.2	15.7	15.0	15.3	15.3	15.4	16.4	18.5
	5%	DB	25.7	26.0	25.0	23.0	21.1	19.1	18.2	18.1	18.8	19.8	21.2	23.4
		MCWB	19.8	19.4	19.3	17.7	16.6	15.0	14.2	14.2	14.9	15.2	16.2	17.8
	10%	DB	24.9	25.1	24.1	22.1	20.2	18.2	17.6	17.2	18.1	19.1	20.8	22.9
		MCWB	19.2	19.1	18.6	17.3	15.9	14.4	13.9	13.7	14.2	14.8	16.0	17.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.1	21.6	21.1	19.9	18.6	16.8	16.2	17.5	17.1	16.6	17.7	20.2
		MCDB	26.3	25.9	25.0	23.7	22.7	19.9	18.9	20.1	19.7	19.7	21.4	24.1
	2%	WB	21.2	20.7	20.3	19.0	17.6	15.9	15.4	15.5	16.0	16.0	17.1	19.4
		MCDB	25.2	25.5	24.6	22.9	21.1	19.1	18.2	18.3	18.8	19.1	20.7	23.2
	5%	WB	20.6	20.1	19.7	18.4	17.0	15.3	14.8	14.6	15.2	15.6	16.6	18.6
		MCDB	24.6	24.7	24.1	22.2	20.1	18.5	17.7	17.4	18.0	18.8	20.3	22.5
	10%	WB	19.8	19.6	19.2	17.7	16.3	14.9	14.2	14.0	14.6	15.2	16.2	18.0
		MCDB	24.0	24.2	23.5	21.4	19.5	17.9	17.0	16.8	17.5	18.5	19.9	22.0
Mean Daily Temperature Range	5% DB	MDBR	5.4	5.8	5.8	5.2	4.5	3.5	3.1	3.1	3.4	3.6	4.3	5.0
		MCDBR	5.8	6.3	6.3	6.0	5.1	4.1	3.6	3.8	3.8	4.0	4.6	5.3
	5% WB	MCWBR	3.1	3.1	3.2	3.3	3.0	2.6	2.5	2.6	2.5	2.5	2.7	3.0
		MCDWR	5.4	5.9	6.1	5.8	5.2	4.1	3.4	3.7	3.7	3.8	4.6	5.4
Clear-Sky Solar Irradiance	taub	taub	0.461	0.453	0.421	0.404	0.373	0.362	0.368	0.367	0.379	0.374	0.377	0.419
		taud	2.202	2.244	2.323	2.317	2.364	2.386	2.370	2.356	2.316	2.332	2.345	2.262
	Ebn at Noon	Ebn at Noon	889	885	888	861	850	841	848	885	913	946	960	928
		Edh at Noon	156	147	131	122	108	101	106	115	129	133	134	147
All-Sky Solar Radiation	RadAvg	RadAvg	8.06	7.56	6.93	5.64	4.43	3.69	3.79	4.52	5.67	6.89	7.84	8.19
	RadStd	RadStd	0.31	0.40	0.28	0.37	0.23	0.17	0.22	0.22	0.27	0.28	0.33	0.20

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page